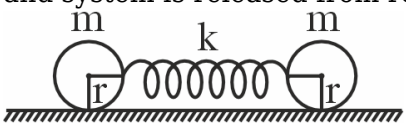
PART-1 : PHYSICS

SECTION-I (i)

1) Two cylinders of equal mass m = 1kg and radius r = 10 cm are connected to opposite ends of a spring having spring constant k = 300 N/m and placed on rough horizontal surface. Spring is compressed by 10 cm and system is released from rest. Assume no slipping ($g = 10\ m/s^2$). Select correct alternatives :-



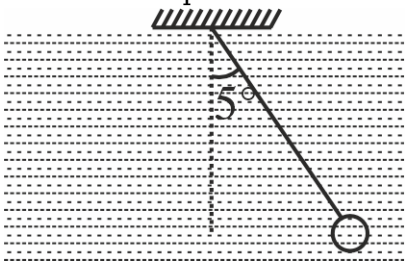
(A) System will perform simple hormonic motion with time period $T = \frac{\pi}{10}s$.

(B) System will perform simple harmonic motion with time period $T = \frac{\pi}{10}\sqrt{\frac{2}{3}}s$.

(C) Minimum coefficient of friction between cylinders and ground should be $\mu = 1$.

(D) Maximum speed of cylinder during motion will be 1 m/s.

2) A spherical bob of density $1000\ kg/m^3$, radius 1cm is connected to a string of length $\ell = 1m$ and other end is fixed. The whole system is submerged in a liquid of density of $500\ kg/m^3$ and viscosity $\eta = \left(\frac{2}{45}\right)$ Pa-s. Pendulum is displaced by 5° from mean position and then released. (neglect moment of inertia of sphere about it's centre of mass) take ($e^{-\pi} \approx 0.04$). Select correct alternatives.



(A) Motion of pendulum is overdamped.

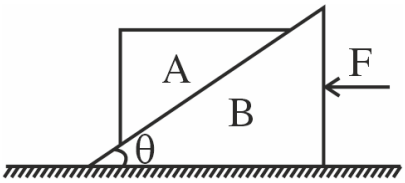
(B) Motion of pendulum is under damped with angular frequency 2rad/s.

(C) Amplitude of oscillation will drop to 0.2° after one oscillation.

(D) Q value of oscillation is $\frac{\sqrt{5}}{2}$.

3) A pair of wedges are located on horizontal surface. The coefficient of friction for every surface is μ. Mass of top wedge A is m, and mass of bottom wedge is M = 2m. A horizontal force F directed to left is applied to bottom wedge, select correct alternative(s). [Take m = 1kg, $g = 10\ m/s^2$)

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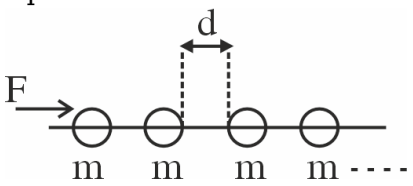
(A) For θ = 37° and μ = 0.8 there will be no slidding between wedges for 0 ≤ F ≤140.25 N

(B) For θ = 37° and μ = 1.3 there will be no slipping between blocks for any value of force F

(C) for θ = 37° and μ = 0.5 there will be no slipping between blocks for 20.45 ≤ F ≤ 75 N

(D) For θ = 37° and μ = 0, there will be no slipping between blocks for F = 22.5 N only

4) Infinite Beads of equal mass are strung at equal distances "d" an a long horizontal rod. The beads are initially at rest but can move without friction. One of the bead is continuously accelerated towards right by constant force F. What are the speed of the accelerated bead, after a long time? (if speed is still variable then find it's mean value)



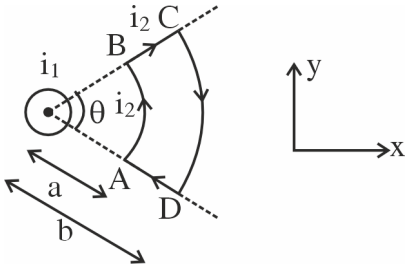
(A) $\sqrt{\frac{Fd}{m}}$, if collisions are perfectly inelastic

(B) $\sqrt{\frac{2Fd}{m}}$, if collisions are perfectly inelastic.

(C) $\sqrt{\frac{Fd}{2m}}$, if collisions are elastic.

(D) $\sqrt{\frac{Fd}{3m}}$, if collision are elastic

5) A long straight wire carry current i_1 = 2A in positive z-direction. Coill ABCD carry current i_2 = 1A. AB and CD are circular arc of radius a = 1 cm and b = 2 cm respectively. Then (Take θ = 74°)



(A) Net force on coil due to wire is zero

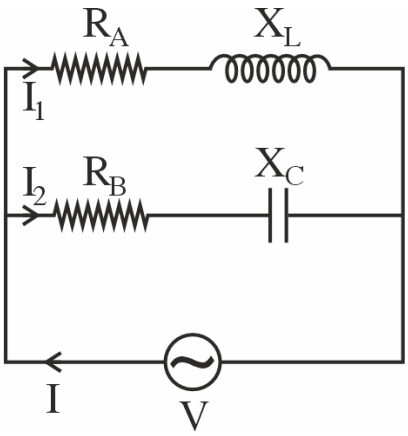
(B) Net force on coil due to wire is 2×10^{-7} N $\hat{k}$.

(C) Net torque on coil due to wire is 4.8×10^{-9} N-m ($\hat{i}$)

(D) Net torque on coil due to wire is zero.

6) In given circuit R_A = 10 Ω, R_B = 10 Ω, $X_L = 10\sqrt{3}\Omega$ and $X_C = \frac{10}{\sqrt{3}}\Omega$. Voltage of alternating source is

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v = 100 volt then

(A) $I_1 = 5A$

(B) $I_2 = 5\sqrt{3}$ A

(C) $I = 5(\sqrt{3} + 1)A$

(D) Total power dissipated is P = 1 kW.

SECTION-I (ii)

Common Content for Question No. 1 to 2

A particle A is fixed at origin and another particle B experiences a force $\vec{F} = \left(+\frac{3\alpha}{r^4} - \frac{2\beta}{r^3}\right)\hat{r}$ due to particle A. $\vec{r}$-represent position vector of particle B with respect to A. Assume potential energy at infinity to be zero. Particle B is in stable equilibrium.

1) Find minimum energy required to increase seperation between A and B to be infinity :-

(A) $-\frac{4\beta^3}{27\alpha^2}$

(B) $+\frac{4\beta^3}{27\alpha^2}$

(C) $-\frac{\beta^2}{4\alpha}$

(D) $+\frac{\beta^2}{4\alpha}$

2) If particle B has mass m then on small disturbance along radial direction it performes oscillation. Find it's time period :-

(A) $\frac{9\pi\alpha^2}{2\beta^2}\sqrt{\frac{m}{\beta}}$

(B) $\frac{9\pi\alpha^2}{2\beta^2}\sqrt{\frac{m}{2\beta}}$

(C) $\frac{9\pi\beta^2}{2\alpha^2}\sqrt{\frac{m}{2\alpha}}$

(D) $\frac{9\pi\beta^2}{2\alpha^2}\sqrt{\frac{m}{\alpha}}$

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Common Content for Question No. 3 to 4

A particle of mass m, charge q is projected from origin with velocity $\vec{v} = v_0\hat{i}$ in a region where uniform magnetic field $\vec{B} = B_0\left(-\hat{k}\right)$ exist. If it also experience a drag force $\vec{F} = -k\vec{v}$, where k is a positive constant and $\vec{v}$ is velocity vector then.

3) Find speed of particle as function of time

(A) $v = v_0\left[e^{-\frac{k}{m}t} + e^{-\frac{qB}{m}t}\right]$

(B) $v = v_0\left[e^{-\frac{k}{m}t}\right]$

(C) $v = v_0 e^{-\frac{qB}{m}t}$

(D) $v = v_0\left(1 - \frac{k}{m}t\right)$

4) Find distance travelled by particle before it's velocity become parallel to the x-axis.

(A) $\frac{mv_0}{k}\left[1 - e^{-\frac{2\pi k}{qB}}\right]$

(B) $\frac{2mv_0}{k}\left[1 - e^{-\frac{2\pi k}{qB}}\right]$

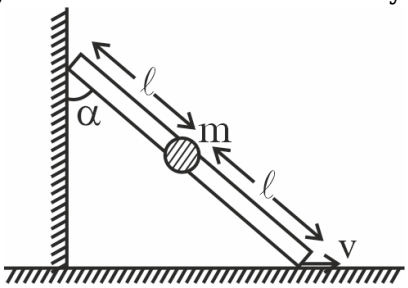
(C) $\frac{mv_0}{qB}\left[1 - e^{-\frac{qB}{2\pi k}}\right]$

(D) $\frac{2mv_0}{qB}\left[1 - e^{-\frac{qB}{2\pi k}}\right]$

SECTION-II (i)

Common Content for Question No. 1 to 2

The length of a weightless rod is 2□. A small sphere of mass m is fixed at distance □ from upper end. The rod rests with it's one end against wall and the other against the floor. The end that rests on the floor is now being moved with constant velocity v away from the wall. Take (v = 2 m/s, □ = 1m, m = 1kg, g = 10 m/s^2)



1) Find the force (in N) applied by sphere on the rod when α = 45°.

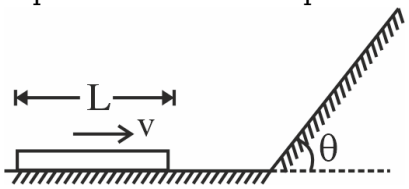
2) Find vertical acceleration (in m/s^2) of sphere when α = 60°.

Common Content for Question No. 3 to 4

A rope of uniform density move under it's own momentum at constant velocity v. It approaches on

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incline at angle 30° and moves all the way up untill half of it's length is on the incline. Total length of rope is 5m. Assume rope remains taut through out process. Neglect any effect collision.

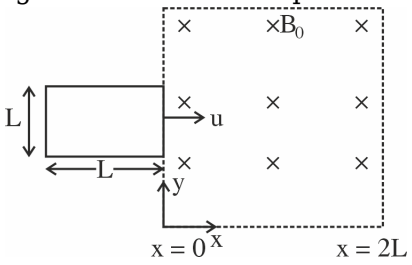


3) Find initial speed (in m/s) of rope :-

4) Find time (in sec) taken by rope to reach maximum height on inclined.

Common Content for Question No. 5 to 6

In gravity free space uniform magnetic field B = 1T exist from x = 0 to x = 2L. A uniform square of length L = 1m, has resistance R = 2Ω is projected along x-axis with same speed u. Mass of square loop is 0.5 kg and at t = 0 it's just entering the field.



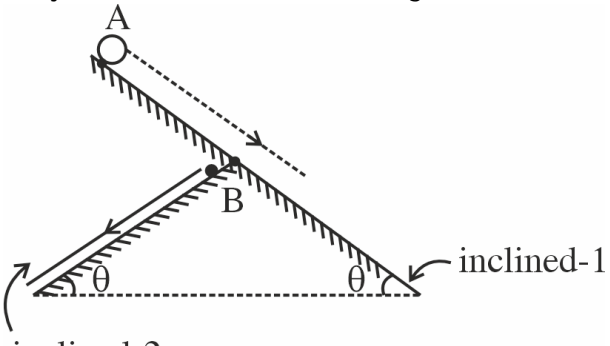
5) Find minimum initial speed (in m/s) of loop so that it just crosses the magnetic field region.

6) If loop was projected with double of minimum speed required to cross the region the find heat (in J) produced in process.

SECTION-II (ii)

1) An object is projected from ground so that it just crosses a 4.8 m high wall at distance 4m from point of projection and hits ground at distance 6m from wall. Initial speed of projectile is $a\sqrt{b}$ m/s then find the value of $\frac{a}{b}$.(a & b are least possible integer) (Take : g = 10 m/s^2)

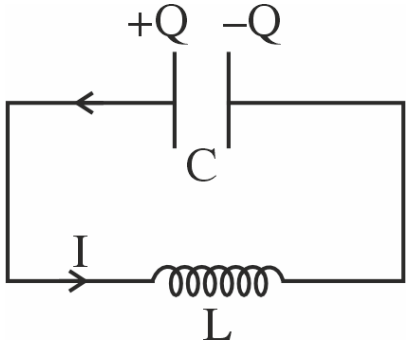
2) Two balls are simultaneously released on two frictionless inclined planes from position A on 1st inclined and from position B on second inclined. The ball pass through a particular horizontal level 12 s and 4s after they where released. How long (in sec) after they were released will they be closest to each other?



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3) In given circuit C = 1μF, Q = 2μC, L = 0.1 mH and i = 0.2 A at given moment. The minimum time after which current in inductor will be maximum is $\frac{a}{b}\pi$ μs. Where a and b are least integer. Find (a +

b).



PART-2 : CHEMISTRY

SECTION-I (i)

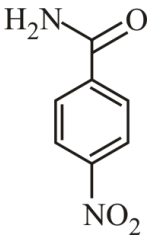
1) Identify the true statement(s)

(A) A catalyst remain chemically unchanged at the end of a reaction
(B) A catalyst may appear in the kinetic rate equation of the reaction
(C) Addition of catalyst will not affect the composition of an equilibrium mixture
(D) Addition of catalyst cannot cause a non-spontaneous reaction to proceed

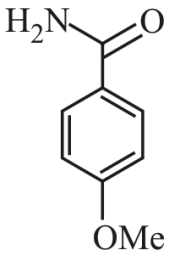
2) Which of the following statements is/are correct?

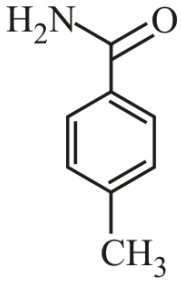
(A) Dislocation of ion from lattice site to interstitial site is called Frenkel defect
(B) Missing of positive and negative ions from their respective position producing holes is called Schottky defect
(C) The presence of ions in the vacant interstitial sites along with lattice point is called interstitial defect
(D) When sodium vapours passed on NaCl crystal, it becomes yellow in colour due to non-stoichiometric defect

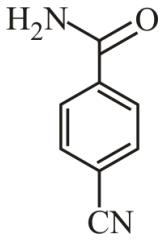
3) Which compound(s) is/are less reactive than benzamide towards Hofmann bromamide reaction?

(A) 

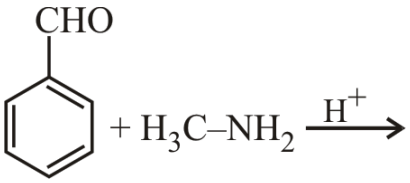
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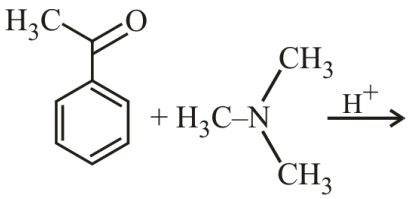
(B) 

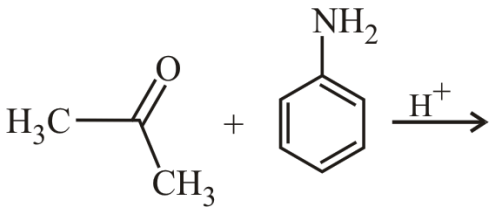
(C) 

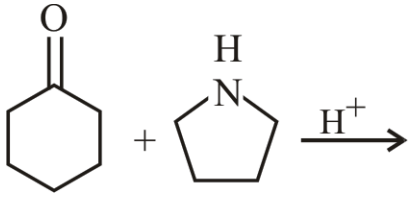
(D) 

4) Which of the following reaction(s) lead to formation of Schiff's base?

(A) 

(B) 

(C) 

(D) 

5) The correct statement(s) related to processes involved in the extraction of metals is(are)

(A) Roasting of Malachite produces Cuprite.

(B) Calcination of Calamine produces Zincite.

(C) Copper pyrites is heated with silica in a reverberatory furnace to remove iron.

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(D) Impure silver is treated with aqueous KCN in the presence of oxygen followed by reduction with zinc metal.

6) The electrochemical extraction of aluminum from bauxite ore involves

(A) the reaction of Al_2O_3 with coke (C) at a temperature > 2500 ºC.

(B) the neutralization of aluminate solution by passing CO_2 gas to precipitate hydrated alumina ($Al_2O_3.3H_2O$)

(C) the dissolution of Al_2O_3 in hot aqueous NaOH.

(D) the electrolysis of Al_2O_3 mixed with Na_3AlF_6 to give Al and CO_2.

SECTION-I (ii)

Common Content for Question No. 1 to 2

Spinel is an important class of oxide having two types of metal ions. A type of spinel have CCP arrangement of O^{2-} ions in which Fe^{2+} cation occupy $1/8^{th}$ of tetrahedral voids and Fe^{3+} cation occupies half of the octahedral voids.

1)

What is formula of spinel ?

(A) Fe_2O_3

(B) Fe_3O_4

(C) FeO

(D) $FeFe_2O_4$

2)

What is the minimum internuclear distance between Fe^{2+} ion & Fe^{3+} ion in the given spinel ? ('a' be the edge of unit cell.)

(A) $\frac{\sqrt{3}a}{4}$

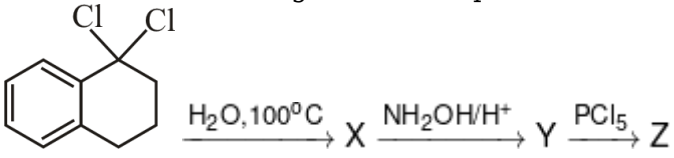
(B) $\frac{a}{2}$

(C) $\frac{a}{\sqrt{2}}$

(D) a

Common Content for Question No. 3 to 4

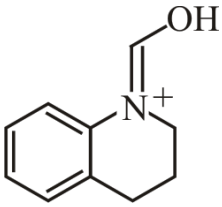
Consider the following reaction sequence :

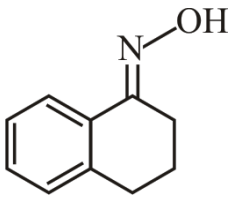


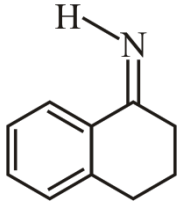
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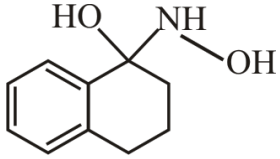
Answer the following questions.

3) The major product Y is :

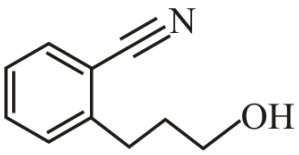
(A) 

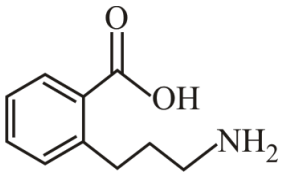
(B) 

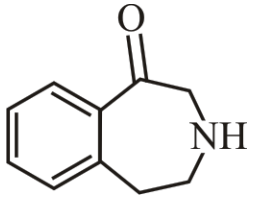
(C) 

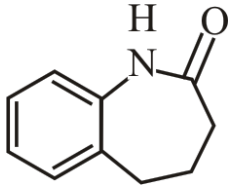
(D) 

4) The major product Z is :

(A) 

(B) 

(C) 

(D) 

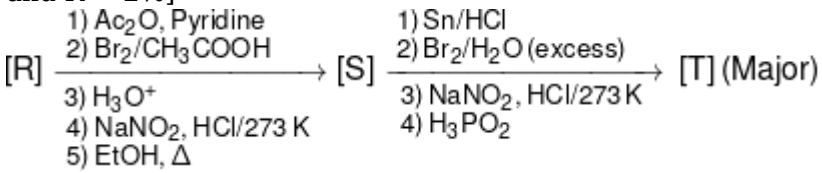
SECTION-II (i)

Common Content for Question No. 1 to 2

Aniline reacts with mixed acid (conc. HNO_3 and conc. H_2SO_4) at 288 K to give [P = 51%, Q = 47%

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and R = 2%]



1) Sum of locant numbers of bromine in [T] according to IUPAC nomenclature is _____

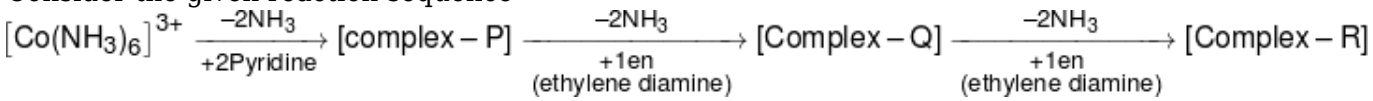
2) The molecular weight (g/mol) of the major product (S) of the following reaction is ____
[Use : Molar mass H = 1, C = 12, N = 14, O = 16, Br = 80]

Common Content for Question No. 3 to 4
In a crystalline structure of diamond also called "diamond cubic", carbon atoms shows FCC structure with four more atoms that are found to be present at the alternate tetrahedral voids.

3) Effective number of C-atoms in diamond cubic unit cell are.....

4) The % void space in diamond is.....

Common Content for Question No. 5 to 6
Consider the given reaction sequence



5) Number of N-Co-N linkages in complex Q are:

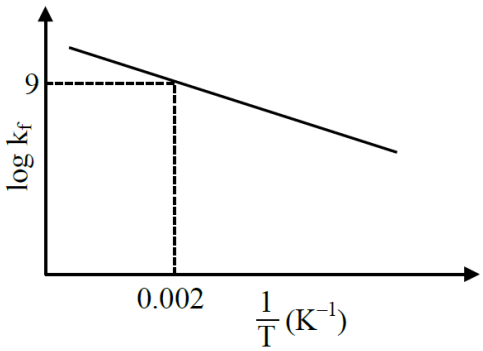
6) Sum of number of Geometrical isomers of Complex - P and Complex-R is:

SECTION-II (ii)

1) An ionic compoud $\left(\overset{+}{X}\ \overset{-}{Y}\right)$ is doped with controlled amount of (MY_2). Number of cation vacancies in 1 mol of XY is found to be 6.02×10^{21}. Atomic mass of X, Y and M are 15,10 and 20 a.m.u, respectively. Assume volume remains unchanged.
If the % decreases in density is x, then the value of 10x is : (Given : $N_A = 6.02 \times 10^{23}$)

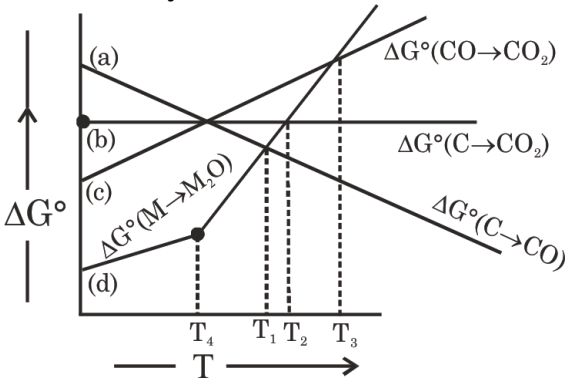
2) The plot of log k_f versus $^1/_T$ for a reversible reaction A (g) $\rightleftharpoons$ P (g) is shown.

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Pre-exponential factors for the forward and backward reactions are 10^{15} s^{-1} and 10^{11} s^{-1}, respectively. If the value of log K for the reaction at 500 K is 6, the value of | log k_b | at 250 K is _____.
[K = equilibrium constant of the reaction, k_f = rate constant of forward reaction, k_b = rate constant of backward reaction]

3) How many statements are correct about following diagram?



A) Carbon can be used to reduce the metal oxide and itself oxidised into carbondioxide at only above the T_3 temperature.
(B) Carbon can be used to reduce the metal oxide and itself oxidised into carbonmonoxide at above T_1 temperature.
(C) Carbon monoxide can reduce the metal oxide to the metal and is oxidised to carbondioxide at above T_3 temperature.
(D) Carbon can be used to achieve the reduction for metal oxide below the T_2 temperature and carbon is oxidised to carbondioxide.
(E) T_4 is the melting point of metal

PART-3 : MATHEMATICS

SECTION-I (i)

1) For the equation $2(2^x - 1)x^2 + (2^{x^2} - 2)x = 2^{x+1} - 2$ for real numbers x

(A) Number of real solutions = 4
(B) Number of real solutions = 3
(C) Sum of real solutions = 0
(D) Sum of real solutions = 2

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2) If S is the set of all real numbers 'x' such that $\frac{x^2-|x|-2}{2|x|-x^2-2} > 2$, then S contains -

(A) $(6, \infty)$

(B) $\left(\frac{2}{3}, 1\right)$

(C) $\left(-1, -\frac{2}{3}\right)$

(D) (–2, –1)

3) A line divides a plane into 2 regions. Two lines divide the plane into maximum 4 regions. If L_n is the maximum number of regions divided by n lines then which of the following is/are true :

(A) $L_{20} = 211$

(B) $L_{10} = 56$

(C) $L_{15} = 121$

(D) $L_{25} = 326$

4) If $i = \sqrt{-1}$ and k is a natural number, then value of $\left|\frac{2^k}{(1+i)^{2k}} + \frac{(1+i)^{2k}}{2^k}\right|$ is -

(A) 0 when k is odd

(B) 2 when k is even

(C) 1 when k is odd

(D) $\frac{1}{2}$ when k is even

5) A bag contains n white and n black balls. Pairs of balls are drawn from the bag without replacement untill the bag is empty. Probability that each draw consists of one white and one black ball is given by P(n), then-

(A) $P(2) = \frac{2}{3}$

(B) $P(6) = \frac{32}{231}$

(C) $P(2) = \frac{1}{3}$

(D) $P(6) = \frac{16}{231}$

6) Let

$N = 2^{200}\left({}^{200}C_0\right)\left({}^{200}C_{100}\right) - 2^{198}\left({}^{200}C_1\right)\left({}^{199}C_{99}\right) + 2^{196}\left({}^{200}C_2\right)\left({}^{198}C_{98}\right) - \ldots\ldots + \left({}^{200}C_{100}\right)\left({}^{100}C_0\right)$

then

(A) N is equal to the coefficient of x^{100} in $(1 - x)^{200}$

(B) The number of zeroes at the end of N is only one.

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(C) $N = 3^{101} \times k$ where k is an integer.

(D) N is equal to the number of ways of distributing 200 different objects among 2 persons equally.

SECTION-I (ii)

Common Content for Question No. 1 to 2

Let $(1 + x + x^2)^{20} = a_0 + a_1x + a_2x^2 + \ldots\ldots + a_{40}x^{40}$. Let $S = \sum_{r=0}^{40} a_r$ and a is unit digit of S. On the basis of above information, answer the following questions :

1) If $\left(\sqrt{2+a} + a\right)^6 = I + f$ where $0 \le f < 1$ and $I \in N$, then the value of I is -

(A) 413

(B) 414

(C) 415

(D) 416

2) The value of a_3 is -

(A) $\frac{3\,\lfloor 20}{\lfloor 18}$

(B) $\frac{4\,\lfloor 20}{\lfloor 18}$

(C) $\frac{\lfloor 20}{\lfloor 17\ \lfloor 3}$

(D) $\frac{\lfloor 20}{\lfloor 18}$

Common Content for Question No. 3 to 4

Let S_n be the sum of infinite terms of geometric series whose first term is n and the common ratio is $\frac{n+1}{2n+1}$ (where n = 1, 2, 3, ...).

3) Value of $S_1S_n + S_2S_{n-1} + \ldots\ldots + S_nS_1$ is -

(A) n

(B) 2n(n + 1)

(C) $\frac{n}{3}(2n^2 + 12n + 13)$

(D) $\frac{n}{3}(2n + 3)^2$

4) Value of $S_1^2 + S_2^2 + \ldots\ldots + S_n^2$ is -

(A) $\frac{n}{2}(2n + 3)^2$

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(B) $n(2n + 3)^2$

(C) $\frac{n}{3}(4n^2 + 12n + 11)$

(D) $(2n + 1)^2$

SECTION-II (i)

Common Content for Question No. 1 to 2

Contents of the Urn-I,II and III are as follows.

A dice is thrown if a prime number appears, then a ball is drawn from Urn-I and if composite number appears, a ball is drawn from Urn-II otherwise a ball is drawn from Urn-III. Given that a white ball is drawn, then

Urn	W	B
I	3	3
II	2	3
III	4	2

1) If P is the probability that Urn-III was selected then [15P] is equal to
(where [.] represents greatest integer function)

2) If another ball is drawn from the same Urn, then the probability that it is a black ball (Given that first drawn ball is not replaced) is q then [10q] is (where [.] represents greatest integer function)

Common Content for Question No. 3 to 4

Roots of $w^n - 1 = 0$ are $1, \alpha, \alpha^2, \alpha^3, \ldots\ldots \alpha^{n-1}$, (where n is prime number and $\alpha = e^{i\frac{2\pi}{n}}$, $i = \sqrt{-1}$)
On the basis of above information, answer the following questions :

3) If n = 7, then the equation whose roots are $\alpha + \alpha^2 + \alpha^4$ and $\alpha^3 + \alpha^5 + \alpha^6$ is $x^2 + bx + c = 0$ then (b + c) is

4) If n = 5, then $(1 + \alpha)(1 + \alpha^2)(1 + \alpha^3)(1 + \alpha^4)$ is equal to -

Common Content for Question No. 5 to 6

Consider 3 alike english books, 5 alike hindi books and 3 different science books.

5) How many different selections can be made so that each selection consists of exactly one book on each subject ?

6) How many different selections can be made so that each selection consists 4 books and atleast one book on each subject ?

SECTION-II (ii)

1) Median of the following observations is

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x	10	30	20	40	60	50
f	5	17	8	13	7	10

2) The variance of 20 observation is 5. If each observation is multiplied by 2, then the new variance of the resulting observations, is

3) The geometric and harmonic means of two positive number x_1 and x_2 are 18 and $16\frac{8}{13}$ respectively. The value of $|x_1 - x_2|$ is -

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ANSWER KEYS

PART-1 : PHYSICS

SECTION-I (i)

Q.	1	2	3	4	5	6
A.	A,C,D	B,C,D	A,B,C,D	A,C	A,C	A,B,D

SECTION-I (ii)

Q.	7	8	9	10
A.	B	B	B	A

SECTION-II (i)

Q.	11	12	13	14	15	16
A.	7.10 to 7.20	8.00	2.50	1.57	2.00	3.00

SECTION-II (ii)

Q.	17	18	19
A.	1	8	7

PART-2 : CHEMISTRY

SECTION-I (i)

Q.	20	21	22	23	24	25
A.	A,B,C,D	A,B,C,D	A,D	A,C	A,B,C	B,C,D

SECTION-I (ii)

Q.	26	27	28	29
A.	B	A	B	D

SECTION-II (i)

Q.	30	31	32	33	34	35
A.	11.00	202.00	8.00	65.96 to 66.00	15.00	4.00

SECTION-II (ii)

Q.	36	37	38
A.	4	5	3

PART-3 : MATHEMATICS

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SECTION-I (i)

Q.	39	40	41	42	43	44
A.	B,C	B,C	A,B,C,D	A,B	A,D	B,C

SECTION-I (ii)

Q.	45	46	47	48
A.	C	B	C	C

SECTION-II (i)

Q.	49	50	51	52	53	54
A.	3.00	5.00	3.00	1.00	3.00	9.00

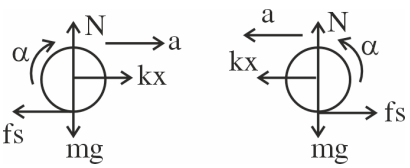
SECTION-II (ii)

Q.	55	56	57
A.	35	20	15

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SOLUTIONS

PART-1 : PHYSICS



1)

kx – fr = ma .....(1)

$frR = \frac{mR^2}{2}\left(\frac{a}{R}\right)$

$fr = \frac{ma}{2}$ .....(2)

$kx = \frac{3}{2}ma$ .....(3)

$\frac{d^2x}{dt^2}$ = relative acceleration = 2a

$\Rightarrow kx = \frac{3}{4}m\left(\frac{d^2x}{dt^2}\right)$

$\Rightarrow \frac{d^2x}{dt^2} = \frac{4k}{3m}x = \omega^2 x$

(a) $T = 2\pi\sqrt{\frac{3m}{4k}} = 2\pi\sqrt{\frac{3\times 1}{4\times 500}} = \frac{2\pi}{20} = \frac{\pi}{10}s$

(c) kx – fs = ma = 2fr

$fr = \frac{kx}{3}$

$(fs)_{max} = \frac{kx_{max}}{3} = \mu mg$

$\mu = \frac{300\times 0.1}{3\times 1\times 10} = 1$

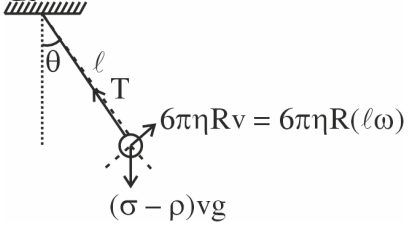
(d) $v_{max} = A\omega = (5cm)\left(\frac{2\pi}{\pi}\times 10\right)$ = 100 cm/s = 1m/s

2)

$\tau = (\sigma - \rho)\left(\frac{4}{3}\pi R^3\right)g\ell\theta - \left(6\pi\eta R\ell^2\right)\omega = -\left[\sigma\left(\frac{4}{3}\pi R^3\right)\ell^2\right]\alpha$

$\alpha + \left(\frac{\sigma-\rho}{\sigma}\right)\frac{g}{\ell}\theta - \frac{9\eta}{2\sigma R^2}\omega = 0$

$\frac{d^2\theta}{dt^2} - 2\gamma\frac{d\theta}{dt} + \omega_0^2\theta = 0$



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(A) $\omega_0 = \sqrt{\left(\frac{\sigma-\rho}{\sigma}\right)\left(\frac{9}{\ell}\right)} = \sqrt{\left(\frac{1000-500}{1000}\right)\left(\frac{10}{1}\right)} = \sqrt{5}$ rad/s

(B) $\ell n\left(\frac{A}{A_0}\right) = -\gamma T = -1\left(\frac{2\pi}{2}\right) = -\pi$

$A = A_0 e^{-\pi} = 5 \times 0.04 = 0.2^\circ$

(C) $Q = \frac{\omega_0}{2\gamma} = \frac{\sqrt{5}}{2(1)} = \frac{\sqrt{5}}{2}$

3) (a) for $\mu > \tan\theta$, no force is required to keep blockes in place but maximum force is given by

$f_{max} = 3mg\left[\frac{2\mu + \left(1-\mu^2\right)\tan\theta}{1-\mu\tan\theta}\right]$

$= 3(1)(10)\left[\frac{2(0.8) + (1-0.64)\left(\frac{3}{4}\right)}{1-0.8\left(\frac{3}{4}\right)}\right] = 140.25N$

(b) for $\mu > \cot\theta$, there will be no slipping between wedges.

(c) for $\mu < \tan\theta$

$f_{min} = 3mg\left(\frac{\left(1+\mu^2\right)\tan\theta}{1+\mu\tan\theta}\right)$

$= 3(1)(10)\left[\frac{(1+0.25)(0.75)}{1+0.5\times 0.75}\right] N = 20.45N$

$F_{max} = 3mg\left[\frac{2\mu + \left(1-\mu^2\right)\tan\theta}{1-\mu\tan\theta}\right]$

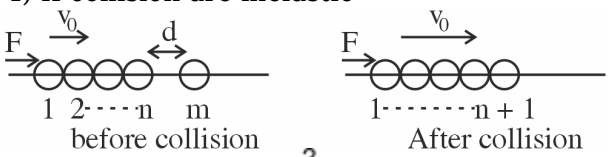
$= 30\left[\frac{1+(1-0.25)(0.75)}{1-0.5\times 0.75}\right] = 75N$

(d) for $\mu = 0$

$f_{min} = f_{max} = 3mg\tan\theta = 30 \times \frac{3}{4}$

$= \frac{45}{2} = 22.5N$

4) If collision are inelastic

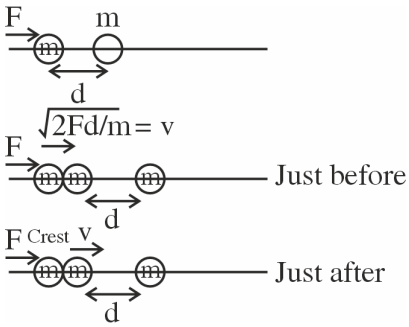


$F = \frac{\Delta p}{\Delta t} = \frac{mv_0}{\left(\frac{d}{v_0}\right)} = \frac{mv_0^2}{d}$

$v_0 = \sqrt{\frac{Fd}{m}}$

If collision in elastic

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$v^2 = u^2 + 2as$

$v^2 = 0 + 2\left(\frac{F}{m}d\right)$

$v = \sqrt{\frac{2Fd}{m}}$

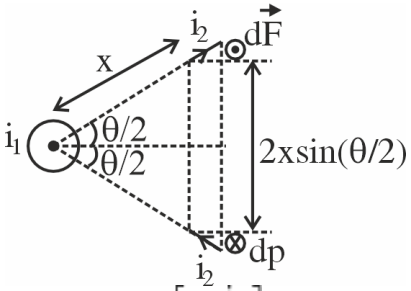
Velocity of 1st bead increases 0 to v then becomes zero than again increases from 0 to v and so on

$\Rightarrow v_{av} = \frac{v}{2} = \sqrt{\frac{Fd}{2m}}$

5) $F_{AB} = F_{CD} = 0$ as $\vec{B} || \vec{d\ell}$

$\vec{F}_{BC} + \vec{F}_{-DA} = 0$ (equal but opposite)

$\Rightarrow$ Net force = 0



$dF = i_2 (dx) \left[\frac{\mu_0 i_1}{2\pi x}\right]$

$d\tau = \frac{\mu_0 i_1 i_2 dx}{2\pi x}\left[2x \sin\left(\frac{\theta}{2}\right)\right]$

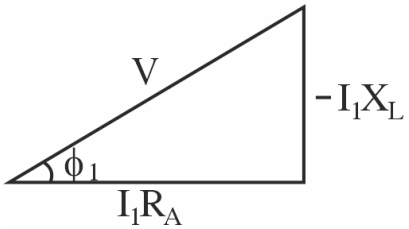
$= \frac{\mu_0 i_1 i_2}{\pi}\sin\left(\frac{\theta}{2}\right)\int_a^b dx$

$\tau = \frac{\mu_0 i_1 i_2}{\pi}\sin\left(\frac{\theta}{2}\right)(a-b)$

$= 4 \times 10^{-7} \times 2 \times \left(\frac{3}{5}\right)(0.01)$

$\tau = 4.8 \times 10^{-9}$ N-m

6)

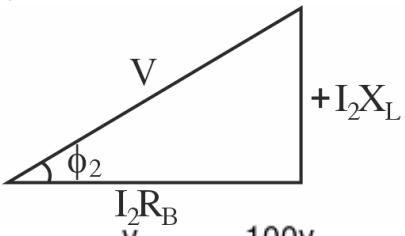


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$$I_1 = \frac{v}{\sqrt{R_A^2 + x_L^2}} = \frac{100v}{20\omega} = 5A \quad \text{.....(1)}$$

$$\tan\phi = -\frac{x_L}{R_A} = -\sqrt{3}$$

$\phi_1 = -60°$ .....(2)



$$I_2 = \frac{v}{\sqrt{R_B^2 + x_C^2}} = \frac{100v}{\frac{20}{\sqrt{3}}\Omega} = 5\sqrt{3}A \quad \text{.....(3)}$$

$$\tan\phi_2 = +\frac{x_C}{R_B} = \frac{1}{\sqrt{3}}$$

$\phi_2 = +30$ .....(ii)

$$I = \sqrt{I_1^2 + I_2^2} = 10A$$

$$P = I_1^2R_A + I_2^2R_B = 250 + 750 = 1000\omega$$

7) $$\vec{F} - = \frac{1}{r^3}\left[+\frac{3\alpha}{r} - 2\beta\right]\hat{r}$$

for equilibrium $\vec{F} = 0$

$$r_0 = \frac{3\alpha}{2\beta} \quad \text{.....(1)}$$

$$U = -\int_{\infty}^{r} \vec{F}.d\vec{r} - = -\left[\int_{\infty}^{r} +\frac{3\alpha}{r^4}dr - \int_{\infty}^{r} \frac{2\beta}{r^3}dr\right]$$

$$U = -\left[+3\alpha\left(\frac{r^{-3}}{-3}\right) - 2\beta\left(\frac{r^{-2}}{-2}\right)\right] = -\frac{\beta}{r^2} + \frac{\alpha}{r^3}$$

$$U = \frac{1}{r_0^2}\left(-\beta + \frac{\alpha}{r_0}\right) = \frac{4\beta^2}{9\alpha^2}\left[\beta + \frac{\alpha}{3\alpha} \times 2\beta\right] = -\frac{4\beta^3}{27\alpha^2}$$

$U_\infty = 0$

$$\Delta E = U_\infty - U^0 = +\frac{4\beta^3}{27\alpha^2}$$

8) $$\vec{F} = \frac{1}{r^4}[+3\alpha - 2\beta r] = 0$$

$$r_0 = \frac{3\alpha}{2\beta}$$

Now if $r = r_0 + \delta r$

$$\vec{F} = \frac{1}{(r_0 + \delta r)^4}[3\alpha - 2\beta(r_0 + \delta r)] \approx \left(-\frac{2\beta}{r_0^4}\right)\delta r$$

$$T = 2\pi\sqrt{\frac{m}{k}} = 2\pi\sqrt{\frac{mr_0^4}{2\beta}}$$

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$= 2\pi r_0^2\sqrt{\frac{m}{2\beta}} = \frac{9\pi\alpha^2}{2\beta^2}\sqrt{\frac{m}{2\beta}}$

9) Drag force will produce tengential accelerate

$a = -\frac{k}{m}v = \frac{dv}{dt}$

$\int_{v_0}^{v}\frac{dv}{v} = -\frac{k}{m}\int_0^t dt$

$v = v_0 e^{-\frac{k}{m}t}$

10) $D = \int vdt = v_0\int_0^t e^{-\frac{k}{m}t}dt$

$D = \frac{mv_0}{k}\left[1-e^{-\frac{k}{m}t}\right]$

$\theta = \frac{qB}{m}t = 2\pi$

$t = \frac{2\pi m}{qB}$

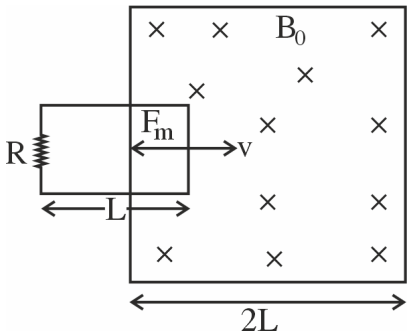
$\Rightarrow D = \frac{mv_0}{k}\left[1-e^{-\frac{2\pi k}{qB}}\right]$

11) $F = m\left(g-\frac{v^2}{\sqrt{2}\ell}\right) = 1\left[10-\frac{4}{\sqrt{2}}\right] = 7.17$ N

12) $a_y = \frac{2v^2}{\ell} = \frac{2(2)^2}{1} = 8$ m/s^2

13) $v = \frac{L}{2}\sqrt{\frac{g\sin\theta}{L}} = \frac{1}{2}\sqrt{gL\sin\theta} = \frac{1}{2}\sqrt{10\times 5\times\frac{1}{2}} = 2.5$ m/s

Alt $\frac{1}{2}mv^2 = \left(\frac{m}{2}\right)g\left(\frac{L}{4}\sin\theta\right)$



16)

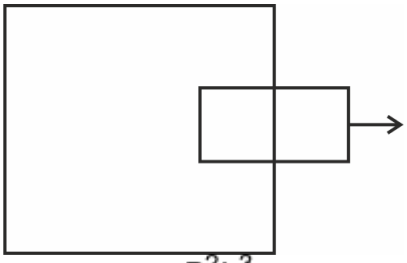
$i = \frac{vBL}{R}$

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$F_m = i\ell B = \frac{B^2L^2}{R}v = -\left(v\frac{dv}{dx}\right)m$

$\int_u^v dv = -\frac{B^2L^2}{mR}\int_0^L dx$

$v - u = \frac{-B^2L^3}{mR}$ .....(1)



(a) $v_f = U - \frac{2B^2L^3}{mR} = 0$

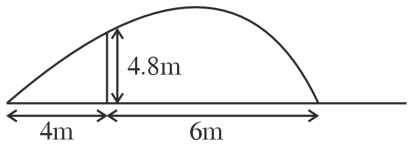
$u = \frac{2B^2L^3}{mR} = \frac{2(1)^2(1)}{(0.5)(2\Omega)} = 2m/s$

(b) If u = 4m/s

$v_f = u - \frac{2B^2L^3}{mR} = 2m/s$

Heat produced = Loss in K.E.

$= \frac{1}{2}m\left(u^2 - v_f^2\right) = \frac{1}{2}(0.5)\left(4^2 - 2^2\right) = 3J$

17) 

$y = \tan\theta\, x\left(1 - \frac{x}{R}\right)$

R = 10 m

and at x = 4m y = 4.8 m

$4.8 = \tan\theta(4)\left(1 - \frac{4}{10}\right) = 2.4\tan\theta$

$\tan\theta = 2 \Rightarrow \sin\theta = \frac{2}{\sqrt{5}}, \cos\theta = \frac{1}{\sqrt{5}}$

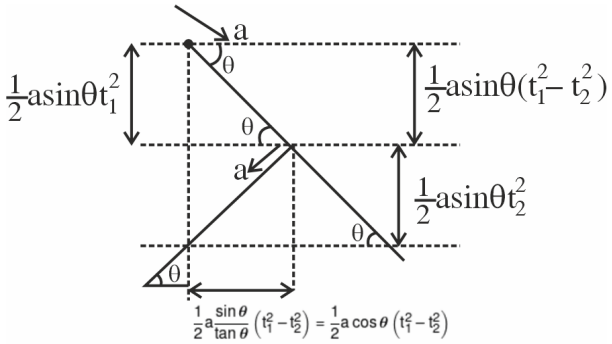
$R = \frac{2u^2\sin\theta\cos\theta}{g}$

$10 = \frac{2u^2\left(\frac{2}{\sqrt{5}}\right)\left(\frac{1}{\sqrt{5}}\right)}{10} = \frac{4u^2}{50}$

$u^2 = 125$

$u = 5\sqrt{5}$ m/s

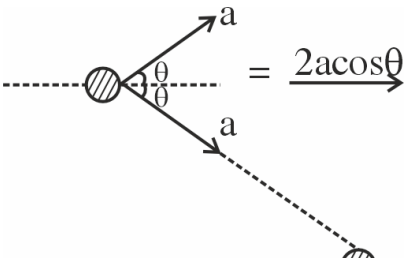
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18)

$$\frac{1}{2}a\frac{\sin\theta}{\tan\theta}\left(t_1^2-t_2^2\right)=\frac{1}{2}a\cos\theta\left(t_1^2-t_2^2\right)$$

Relative acceleration



$$t=\sqrt{\frac{t_1^2-t_2^2}{2}}=\sqrt{\frac{12^2-4^2}{2}}=8s$$

19) $$\omega=\frac{1}{\sqrt{LC}}=\frac{1}{\sqrt{10^{-4}\times10^{-6}}}=10^5 rad/s$$

$$\frac{1}{2}Li_{max}^2=\frac{1}{2}Li^2+\frac{Q_0^2}{2C}$$

$$i_{max}=\sqrt{i^2+\frac{Q^2}{LC}}=\sqrt{0.04+\frac{4\times10^{-12}}{10^{-10}}}$$

$i_{max} = 0.2\sqrt{2}A$

as at t = 0 i = $\frac{i_{max}}{\sqrt{2}}$, time taken to reach maximum will be $\frac{T}{8}=\frac{2\pi}{8\omega}$

$$=\frac{\pi}{4\times10^5}s=\frac{10\pi}{4}\mu s=\frac{5}{2}\pi\mu s$$

PART-2 : CHEMISTRY

26)

$Fe^{3+} = 4\times\frac{1}{2}=2$

$Fe^{2+} = 8\times\frac{1}{8}=1$

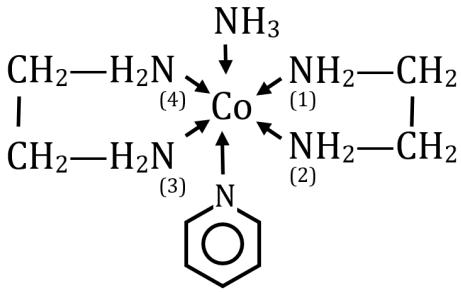
$O^{2-} = 4$

Formula = Fe_3O_4

27)

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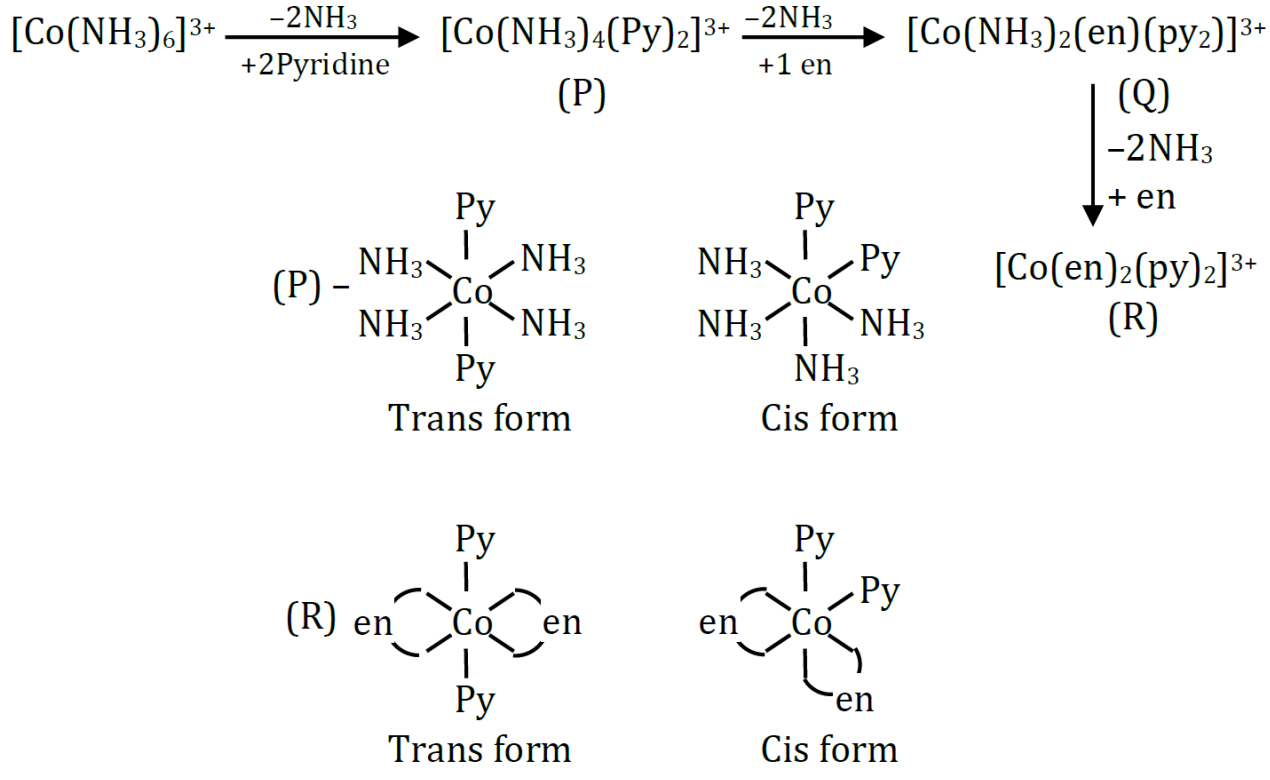
Distance between octahedral & tetrahedral void = $\frac{\sqrt{3}a}{4}$



34)

NH_3 — Co — $NH_{2(1)}$
NH_3 — Co — $NH_{2(2)}$
NH_3 — Co — Py
NH_3 — Co — $NH_{2(3)}$
NH_3 — Co — $NH_{2(4)}$
$NH_{2(1)}$ — Co — $NH_{2(2)}$
$NH_{2(1)}$ — Co — $NH_{2(3)}$
$NH_{2(1)}$ — Co — $NH_{2(4)}$
$NH_{2(2)}$ — Co — $NH_{2(3)}$
$NH_{2(2)}$ — Co — $NH_{2(4)}$
$NH_{2(1)}$ — Co — Py
$NH_{2(2)}$ — Co — Py
$NH_{2(3)}$ — Co — $NH_{2(4)}$
$NH_{2(3)}$ — Co — Py
$NH_{2(4)}$ — Co — Py

35)



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36) Number of moles of cation vacancy

$= \frac{6.02 \times 10^{21}}{6.02 \times 10^{23}} = 10^{-2}\,\text{mole}$

Number of moles of M^{+2} ions = 10^{-2} mole added

decreases in density $= \frac{30 \times 10^{-2} - 20 \times 10^{-2}}{25} \times 100\% = 0.4\%$ = x%

$\Rightarrow$ 10x = 4

37) For reaction A(g) $\rightleftharpoons$ P(g)

$\log k_f = \frac{-E_f}{2.303\,RT} + \log A_f$ [Arrhenius equation for forward reaction]

From plot when, $\frac{1}{T} = 0.002$, $\log k_f = 9$

$\Rightarrow \quad 9 = \frac{-E_f}{2.303\,R}(0.002) + \log (A_f)$

Given : $A_f = 10^{15}\ s^{-1}$

$\Rightarrow \quad 9 = \frac{-E_f}{2.303\,R}(0.002) + 15$

$\Rightarrow \quad \frac{E_f}{2.303R} = \frac{6}{0.002} = 3000$

Now, $K = \frac{k_f}{k_b} = \frac{A_f}{A_b} e^{-(E_f - E_b)/RT}$

$\log K = -\frac{1}{2.303}\frac{(E_f - E_b)}{RT} + \log\left(\frac{10^{15}}{10^{11}}\right)$

At 500 K

$\Rightarrow \quad 6 = \frac{-(E_f - E_b)}{500R\,(2.303)} + 4$

$\Rightarrow \quad (1000\ R)\,(2.303) = E_b - E_f$

$\Rightarrow \quad (1000\ R)\,(2.303) = E_b - 3000\,(2.303\ R)$

$\Rightarrow \quad E_b = 4000\ R\,(2.303) \quad \ldots\ldots\ldots (1)$

Now $k_b = A_b e^{-E_b/RT}$

$\Rightarrow \log k_b = \frac{-E_b}{2.303\,RT} + \log A_b$

At 250 K

$\Rightarrow \quad \log k_b = -\frac{4000}{250} + \log (10^{11})$ [From equation (1)]

= –16 + 11 = –5

$|\log k_b| = 5$

PART-3 : MATHEMATICS

39) Rearranging terms by power of 2 yields

$2^{x^2}x + 2^{x+1}(x^2 - 1) - 2(x^2 + x - 1) = 0 \qquad \ldots(1)$

Setting $y = x^2 - 1$ and dividing by 2 on the both sides, (1) becomes

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$2^y x + 2^x y - (x + y) = 0$
or $x(2^y - 1) + y(2^x - 1) = 0$ ...(2)
Since $f(x) = 2^x - 1$ and x always have the same sign,
$x(2^y - 1) \cdot y(2^x - 1) \geqslant 0$
Hence if the terms on the left-hand side of (2) are nonzero, they must have the same sign, which in term implies that their sum is not equal to 0.
Therefore (2) is true if and only if $x = 0$ or $y = 0$, which leads to solutions $x = -1,\ 0$ and 1

40) $\dfrac{x^2 - |x| - 2}{2|x| - x^2 - 2} - 2 > 0$

$\dfrac{x^2 - |x| - 2 - 4|x| + 2x^2 + 4}{2|x| - x^2 - 2} > 0$

$\dfrac{3x^2 - 5|x| + 2}{-x^2 + 2|x| - 2} > 0$

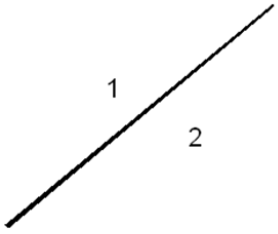
$\dfrac{3|x|^2 - 5|x| + 2}{|x|^2 - 2|x| + 2} < 0$

$(3|x| - 2)(|x| - 1) < 0$

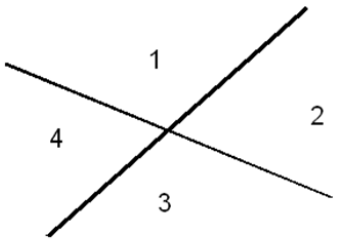
$\dfrac{2}{3} < |x| < 1$

Hence $\dfrac{2}{3} < x < 1$ or $-1 < x < -\dfrac{2}{3}$

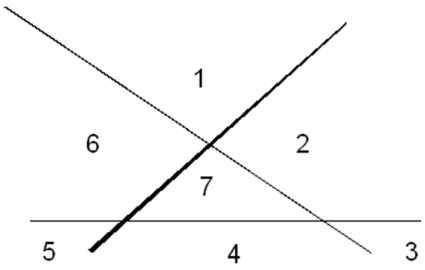
41) Clearly the number of regions is maximized when no two lines are parallel and no three and concurrent. Easy to see that $L_1 = 2$, $L_2 = 4$ and $L_3 = 7$. For the general case, we observe that when we add the nth line, this new line is cut by the previous n – 1 lines in n – 1 points and hence the new line is divided into n segments. Each of these segments divides a previous region into 2 parts and hence the number of new regions



$L_1 = 2$



$L_2 = 4$



$L_3 = 7$

Lines dividing a plane

added when we introduce the nth line is n. Thus we have the recurrence relation
$L_n = L_{n-1} + n.$ Hence $L_4 = 11,\ L_5 = 16, \ldots$
$L_n - L_{n-1} = n \quad L_{n-1} - L_{n-2} = n - 1$

....

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Adding, we get

$L_n - L_1 = n + (n-1) + (n-2) + \ldots + 2$

$= \frac{n(n+1)}{2} - 1$

Thus $L_n = \frac{n(n+1)}{2} - 1 + 2$

$= \frac{n(n+1)}{2} + 1$

$= \frac{1}{2}(n^2 + n + 2)$

$L_{10} = 56$

$L_{15} = 121$

$L_{20} = 211$

$L_{25} = 326$

42) $\left| \frac{2^k}{\left((1+i)^2\right)^k} + \frac{\left((1+i)^2\right)^k}{2^k} \right| = \left| \frac{2^k}{(2i)^k} + \frac{(2i)^k}{2^k} \right|$

$= \left| \frac{1}{i^k} + i^k \right| = \left| \frac{i^k}{i^{2k}} + i^k \right| = i^k \left(\frac{1}{(-1)^k} + 1 \right)$

$= \left| i^k \left((-1)^k + 1 \right) \right|$

If k is odd, 0

If k is even, 2

43)

considering n white and n black balls,

$n(S) = {}^{2n}C_2 . {}^{2n-2}C_2 . {}^{2n-4}C_2 \ldots\ldots {}^4C_2 . {}^2C_2$

$n(S) = \frac{(2n)!}{2^n}$

$n(E) = ({}^nC_1 . {}^nC_1)({}^{n-1}C_1 . {}^{n-1}C_1) \ldots\ldots . ({}^1C_1 . {}^1C_1)$

$= (n!)^2$

$P = \frac{n(E)}{n(S)} = \frac{(n!)^2}{(2n)!} . 2^n = \frac{2^n}{{}^{2n}C_n}$

$P(2) = \frac{2^2}{{}^4C_2} = \frac{2}{3}; P(6) = \frac{16}{231}$

44) $N = \sum_{r=0}^{100} (-1)^r . 4^{100-r} . {}^{200}C_r . {}^{200-r}C_{100-r}$

$= \sum_{r=0}^{100} (-1)^r . 4^{100-r} . \frac{\lfloor 200}{\lfloor r \ \lfloor 200-r}} . \frac{\lfloor 200-r}{\lfloor 100-r \ \lfloor 100}} . \frac{\lfloor 100}{\lfloor 100}$

$= {}^{200}C_{100} \sum_{r=0}^{100} (-1)^r . 4^{100-r} . {}^{100}C_r$

$N = 3^{100} . {}^{200}C_{100}$

45) a = 1, $\left(\sqrt{3}+1\right)^6 = I + f$

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$\left(\sqrt{3}-1\right)^6 = f'$

$I + f + f' = \left(\sqrt{3}+1\right)^6 + \left(\sqrt{3}-1\right)^6$

$= 416$

$\because\ 0 \le f < 1,\ 0 \le f' < 1 \Rightarrow 0 \le f + f' \le 2$

$\Rightarrow f + f' = 1$

$\Rightarrow I = 415$

46) $a = 1, \left(\sqrt{3}+1\right)^6 = I + f$

$\left(\sqrt{3}-1\right)^6 = f'$

$I + f + f' = \left(\sqrt{3}+1\right)^6 + \left(\sqrt{3}-1\right)^6$

$= 416$

$\because\ 0 \le f < 1,\ 0 \le f' < 1 \Rightarrow 0 \le f + f' \le 2$

$\Rightarrow f + f' = 1$

$\Rightarrow I = 415$

47) $S_1 = 3, S_2 = 5, S_3 = 7, \ldots S_{n-1} = 2n-1, S_n = 2n+1$

$S_1S_n + S_2S_{n-1} + \ldots + S_nS_1$

$\Rightarrow 3.(2n + 1) + 5.(2n - 1) + \ldots + (2n + 1).3$

$T_r = (2r + 1)(2n + 3 - 2r)$

$T_r = 2n(2r + 1) + 4r - 4r^2 + 3$

$$\sum_{r=1}^{n} T_r = \frac{n}{3}[2n^2 + 12n + 13]$$

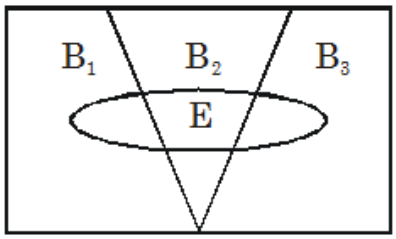
48) Also $S_1^2 + S_2^2 + \ldots + S_n^2$

$3^2 + 5^2 + \ldots + (2n + 1)^2$

$T_r = (2r + 1)^2$

$$\sum T_r = \frac{n}{3}\left[4n^2 + 12n + 11\right]$$

49)



E : A white ball has been drawn.

B_1 : Prime number appears on dice.

B_2 : Composite number appears on dice.

B_3 : a number which is neither prime nor composite appears on dice.

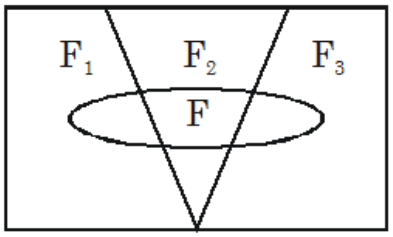
$$\because\ P(B_3/E) = \frac{P(B_3).P(E/B_3)}{P(B_1).P(E/B_1) + P(B_2).P(E)(B_2) + P(B_3).P(E)(B_3)}$$

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$$= \frac{\frac{1}{6}\cdot\frac{4}{6}}{\frac{3}{6}\cdot\frac{3}{6}+\frac{2}{6}\cdot\frac{2}{5}+\frac{1}{6}\cdot\frac{4}{6}} = \frac{20}{89}$$

50)

Let $P(B_3/E) = F_1$



$$P(B_2/E) = \frac{\frac{2}{6}\cdot\frac{2}{5}}{\frac{3}{6}\cdot\frac{3}{6}+\frac{2}{6}\cdot\frac{2}{5}+\frac{1}{6}\cdot\frac{4}{6}} = \frac{24}{89} = P(F_2)$$

&

$$P(B_1/E) = \frac{\frac{3}{6}\cdot\frac{3}{6}}{\frac{3}{6}\cdot\frac{3}{6}+\frac{2}{6}\cdot\frac{2}{5}+\frac{1}{6}\cdot\frac{4}{6}} = \frac{45}{89} = P(F_3)$$

F : Second drawn ball from same urn is a black ball

$\Rightarrow$ $P(F) = P(F \cap F_1) + P(F \cap F_2) + P(F \cap F_3)$

$$= \frac{20}{89}\cdot\frac{2}{5}+\frac{24}{89}\cdot\frac{3}{4}+\frac{45}{89}\cdot\frac{3}{5} = \frac{53}{89}$$

51)

$n = 7$, $\alpha^7 = 1$ and $1 + \alpha + \alpha^2 + \ldots\ldots \alpha^6 = 0$

Sum of the roots $= \alpha + \alpha^2 + \alpha^4 + \alpha^3 + \alpha^5 + \alpha^6 = -1$

Product of the roots $= (\alpha + \alpha^2 + \alpha^4)(\alpha^3 + \alpha^5 + \alpha^6)$

$= \alpha^4 + \alpha^6 + \alpha^7 + \alpha^5 + \alpha^7 + \alpha^8 + \alpha^7 + \alpha^9 + \alpha^{10}$

$= 2 + (1 + \alpha + \alpha^2 + \alpha^3 + \alpha^4 + \alpha^5 + \alpha^6) = 2$

So the required equation is

$x^2 - (-1)x + 2 = 0 \Rightarrow x^2 + x + 2 = 0$

52)

$n = 5$, $\alpha^5 = 1$

and $1 + \alpha + \alpha^2 + \alpha^3 + \alpha^4 = 0$

$(1 + \alpha)(1 + \alpha^2)(1 + \alpha^3)(1 + \alpha^4) = (1 + \alpha + \alpha^2 + \alpha^3)(1 + \alpha^3 + \alpha^4 + \alpha^7)$

$= (-\alpha^4)(1 + \alpha^2 + \alpha^3 + \alpha^4) = (-\alpha^4)(-\alpha) = a^5 = 1$

53) EEE, HHHHH, S_1, S_2, S_3

1E & 1H & 1 science book

$1 \times 1 \times {}^3C_1 = 3$

54) EEE, HHHHH, S_1, S_2, S_3

2E & 1H & IS or 1E & 2H & IS OR

1E & 1H & 2S

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3 + 3 + 3
9

55) Median of the distribution

x	10	20	30	40	50	60
f	5	8	17	13	10	7
C.F.	5	13	30	43	53	60

$\frac{N}{2} = 30$ hence median $= \frac{30+40}{2} = 35$

56) $\frac{1}{20}\sum_{i=1}^{20}(x_i - \bar{x})^2 = 5$

$\sum_{i=1}^{20}(x_i - \bar{x})^2 = 100$

new observations are $2x_1, 2x_2, \ldots\ldots, 2x_{20}$

Their mean $= \bar{x}_1 = \frac{2(x_1 + x_2 + \ldots + x_{20})}{20} = 2\bar{x}$

Now, variance $= \frac{1}{20}\sum_{i=1}^{20}(2x_i - 2\bar{x})^2$

$= \frac{1}{20} \times 4 \sum_{i=1}^{20}(x_i - \bar{x})^2 = \frac{1}{20} \times 4 \times 100 = 20$

57) $x_1x_2 = 18^2 = 12.27$

$\frac{2x_1x_2}{x_1 + x_2} = \frac{216}{13}$

$\Rightarrow 39 = 27 + 12$

$\Rightarrow |x_1 - x_2| = 15$

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